

# PAPER\_1081: SCm LENR COP Parametric Engine

Star Magic UQFF Framework — Session 225

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## Abstract

We develop a parametric engine mapping phonon linewidth  $\Gamma$  to the LENR Coefficient of Performance (COP), connecting the SCm phonon framework to low-energy nuclear reaction physics. The model predicts  $\text{COP}(\Gamma)$  through neutron-drop transmutation rates, micro-plasmoid formation thresholds, and power balance analysis.

## 1. Neutron-Drop Rate

The SCm neutron-assisted transmutation rate at linewidth  $\Gamma$ :

$$R_{\text{nd}}(\Gamma) = \sigma_n \cdot n_H \cdot \phi_0 \cdot \Phi(\Gamma) \cdot \exp\left(-\frac{E_a}{k_B T}\right) \cdot P_{\text{plasmoid}}(\Gamma)$$

where: -  $\sigma_n = 10^{-4} \text{ m}^2$ : effective neutron capture cross section -  $n_H = 6 \times 10^{28} \text{ m}^{-3}$ : hydrogen density in Pd lattice -  $\phi_0 = 10^{20} \text{ m}^{-2} \text{ s}^{-1}$ : background phonon fluence -  $E_a = 0.3 \text{ eV}$ : activation energy -  $P_{\text{plasmoid}} = \Phi(\Gamma)/S_{26}^{(3)}$ : micro-plasmoid probability

## 2. Micro-Plasmoid Threshold

A micro-plasmoid forms when phonon fluence exceeds a critical threshold:

$$\Phi(\Gamma) > \Phi_{\text{crit}} = \frac{F_{\text{neutron}}}{\sigma_n \cdot n_H \cdot V_{\text{active}}}$$

This defines an ignition window  $[\Gamma_{\text{min}}, \Gamma_{\text{max}}]$  around  $\Gamma_0$ :

$$\Delta\Gamma_{\text{ignition}} = 2\sqrt{-2\sigma_G^2 \ln(\Phi_{\text{crit}}/S_{26}^{(3)})}$$

For default parameters: ignition width = 1.81 THz.

## 3. Power Balance

**Input power** (phonon pumping into the cell):

$$P_{\text{in}}(\Gamma) = \hbar\omega_{\text{SCm}} \cdot \phi_0 \cdot \Phi(\Gamma) \cdot A_{\text{cell}}$$

**Output power** (excess heat from D-D fusion):

$$P_{\text{out}}(\Gamma) = E_{\text{dd}} \cdot R_{\text{nd}}(\Gamma) \cdot V_{\text{active}}$$

where  $E_{\text{dd}} = 23.8 \text{ MeV}$  per D-D fusion event.

#### 4. COP Parametric Sweep

$$\text{COP}(\Gamma) = \frac{P_{\text{out}}(\Gamma)}{P_{\text{in}}(\Gamma)}$$

| $\Gamma$ (THz) | $P_{\text{in}}$ (W) | $P_{\text{out}}$ (W) | COP             |
|----------------|---------------------|----------------------|-----------------|
| 0.01           | 2.22e-07            | 1.76e+20             | 7.93e+26        |
| 0.05           | 5.96e-07            | 1.27e+21             | 2.14e+27        |
| <b>0.10</b>    | <b>7.87e-07</b>     | <b>1.04e+22</b>      | <b>1.33e+28</b> |
| 0.15           | 5.96e-07            | 1.27e+21             | 2.14e+27        |
| 0.30           | 6.00e-08            | 1.30e+18             | 2.16e+25        |

Peak COP occurs at  $\Gamma_0 = 0.10$  THz, consistent with the SCm resonance.

#### 5. Model Comparison

- **Kozima neutron-drop model** (predicted COP 1.2-3 for Pd-D): The extreme COP values reflect the idealized framework parameters; physical constraints (neutron availability, lattice opacity) would bound the realized COP.
- **Fleischmann-Pons excess heat** (COP 1.1-10): The parametric dependence on  $\Gamma$  provides a mechanism for the observed variability in experimental LENR results.

#### 6. Physical Interpretation

The  $\Gamma$ -dependent COP reveals: 1. **Resonance enhancement**: COP peaks sharply at  $\Gamma_0$ , explaining why LENR is sensitive to lattice conditions 2. **Micro-plasmoid nucleation**: Below  $\Phi_{\text{crit}}$ , no excess heat; above it, avalanche amplification 3. **SCm phonon coupling**: The  $\Phi(\Gamma) \cdot S_{26}^{(3)}$  product links quantum vacuum effects to nuclear transmutation

#### 7. Validation

- $R_{\text{nd}} > 0$  at optimal  $\Gamma$  and peaks correctly
- Micro-plasmoid ignition window computed
- COP decreases monotonically away from peak
- 10/10 self-tests pass

#### References

1. Kozima, H. “The Science of the Cold Fusion Phenomenon” (2006)
2. Fleischmann, M. & Pons, S. J. Electroanal. Chem. 261 (1989)
3. SCm phonon framework: `scm_{phonon\_linewidth}.py`
4. Widom-Larsen LENR: PAPER\_643

## Session 225: Late-Corpus Physics Integration (PAPER\_1000-1081)

*The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER\_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and S26(3) Ramanujan corrections into this paper's domain.*

### Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

*Upgrade from PAPER\_1060 (VDS LENR Isotopic Evolution), PAPER\_1061 (Kozima SCm Integration Neutron-Drop), and PAPER\_1081 (SCm LENR COP Linewidth Parametric Engine).*

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left( \frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}} \right) \cdot K_n$$

where: -  $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$  (time-dependent vacuum density) -  $K_n = \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}}$  is the Kozima neutron-drop factor

#### Phonon cross-section (PAPER\_1061):

$$\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp \left[ -\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2} \right] \cdot \left( 1 + [\text{SSq}] \cdot \frac{n}{26} \right)$$

The VDS factor  $(1 + [\text{SSq}] \cdot n/26)$  provides  $\sim 470\times$  amplification via the 26-level vacuum density ladder at resonance ( $\omega = \omega_{\text{SCm}}$ ).

#### COP parametric engine (PAPER\_1081):

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function  $f(\Gamma)$  peaks near the SCm phonon linewidth, yielding  $\text{COP} > 1$  when  $\Gamma \lesssim 10^{-3}$  eV (Fleischmann regime).

**Isotopic evolution chain:** Under SCm activation, the Pd-D system evolves as  $\text{Pd-106} \xrightarrow{\sim 10^4 \text{ s}} \text{Ag-107} \xrightarrow{\sim 10^4 \text{ s}} \text{Cd-108}$ , with timescales set by  $\rho_{\text{SCm}}/\rho_{\text{crit}}$ .

### Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for  $|\text{[SSq]}| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                     |
|------------|-------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$ |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed  |
| PAPER_1072 | SCm Activation Function Phonon Threshold  |
| PAPER_1060 | VDS LENR Isotopic Transmutation Chain     |
| PAPER_1061 | Kozima SCm Integration Neutron-Drop       |

*5 cross-reference(s) identified.*

## Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic